MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP(HONOURS) Sample Question Paper
(2024 Admission Onwards)
IV Semester - MG4SECSTA200 - Introduction to Spreadsheet and LaTeX Typing

Time:2 Hrs. Max. Marks: 50

Part A
Answer any SEVEN questions.
Each question carries 2 marks

1. What is a spreadsheet? [U][CO1]
2. Calculate the average of numbers 10, 20, 30, 40 using spreadsheet formula. [U][CO5]
3. Define VLOOKUP. [A][CO2]
4. Name any two spreadsheet functions. [U][CO1]
5. What is pivot table? [Ap][CO3]
6. Write steps to create a bar chart in spreadsheet. [A][CO4]
7. What is the purpose of a

   `\begin{document}`?

   [C][CO7]
8. Write LaTeX code to type the equation

$$a^2 + b^2 = c^2$$

   [C][CO7]
9. Type a simple table in LaTeX with two columns and two rows.[C][CO7]
10. What is Beamer used for? [U][CO6]

$(7 \times 2 = 14)$

Part B
Answer any FOUR questions.
Each carries 6 marks

11. Write the Programm of Standard Deviation using spreadsheets.[A][CO4]

12. Explain (a) Sorting (b) Filtering (c) Freeze panes [A][CO2]

13. Define HLOOKUP function and syntax. [A][CO2]

14. Write the LaTeX code for the following matrix:

$$\begin{pmatrix} 1 & 3 & 2 \\ 3 & 4 & 6 \\ 5 & 7 & 8 \end{pmatrix}$$

[C][CO7]

15. Check the following document and write the out put

```
 The pc indices under normality assumption are defined as follows \\
\begin{equation}\label{c11}
C_{p}=\frac{(USL-LSL)}{6\sigma},
\edn{equation}
where USL is the upper specification limit,
 LSL is the lower specification limit and $\sigma$ is the sd.
\begin{eqauation}
C_{pk}=min\left[\frac{USL-\mu}{3\sigma},\frac{\mu-LSL}{3\sigma}\right],
\end{eqauation}
```

[C][CO7]

16. Write the LaTeX code for the following table:

X:	2.5-7.5	7.5- 12.5	12.5-17.5	17.5-22.5
F:	12	28	65	121

. [C][CO7]

$(4 \times 6 = 24)$

Part C
Answer any ONE questions.
Each carries 12 marks

17. Write the program of any five spreadsheet functions. [A][CO2]

18. Write LaTeX code to produce the following system of equations:

$$\begin{cases} x + y + z = 1 \\ x + y - z = 1 \\ x - y + z = 1 \end{cases}$$

can be written as

$$\begin{pmatrix} 1 & -1 & 1 \\ 1 & 1 & -1 \\ -1 & 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

Here the matrix

$$\begin{pmatrix} 1 & -1 & 1 \\ 1 & 1 & -1 \\ -1 & 1 & 1 \end{pmatrix}$$

is invertible [E][CO7]

$(1 \times 12 = 12)$